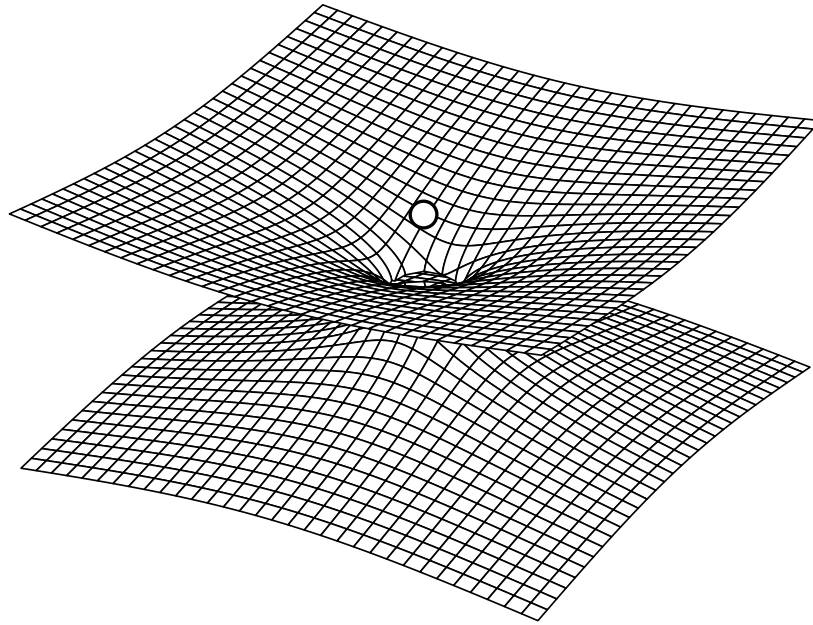




Lecture Notes on General Relativity

PHYSICS 225

VERCIL JUAN
IV - BS Physics

1st Semester, AY 2026-2027

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2}g^{\rho\sigma}(\partial_{g\sigma\nu} + \partial_{\nu}g_{\sigma\mu} - \partial_{\sigma}g_{\mu\nu})$$

$$\frac{d^2x^{\sigma}}{d\lambda^2} + \Gamma_{\mu\nu}^{\sigma}\frac{dx^{\mu}}{d\lambda}\frac{dx^{\nu}}{d\lambda} = 0$$

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$$

Physics 225 - General Relativity 1

Course Information

Course Description

Description: Manifolds, modern differential geometry and tensor analysis; basic principles of general relativity; Einstein field equations and their mathematical properties; exact solutions; linearized theory; variational principles and conservation laws; equations of motion; gravitational waves; experimental tests.

Credits: 3.0

Instructor Details

Instructor: Dr. Francis Ian Vega

E-mail: `ivega.upd.edu.ph`

Lecture Hours: WF, 10:00 - 11:30 AM

Course Requirements

Problem Sets Problem sets will be assigned throughout the semester, ideally one per week. These will be submitted electronically using a Google Form that I will be sending out close to the deadline. From each problem set a random selection of items will be graded. It is your responsibility to ensure that your work is neat and legible. If we cannot understand your work then you will not get any credit. These problem sets will make up the 30

Written Exam There will be two written exams in the course. Each exam will constitute 30% of the final grade. These are in-class, open-notes exams. You will have to designate one notebook as your 225 notebook at the start of the semester. Only this notebook may be used to help you during the exams.

Oral Exam This will be scheduled during or shortly after Finals Week. This will make up the remaining 10 of your final grade.

Main References

We shall not have a main textbook for the class. I used Schutz, supplemented by the first edition of d’Inverno, when I first started learning GR. I think they are both very good texts. However, I have my own way of presenting GR that will not always follow the structure or presentation of any single book. My approach is closest in style to the treatments of Liang+Zhou and Wald. There are numerous sources you can refer to. I highly recommend that you look at different texts to help understand my lectures better. Here are some good books I’ve used on numerous occasions.

For General Relativity

1. R d’Inverno, J Vickers, *Introducing Einstein’s Relativity: A Deeper Understanding*, 2022
2. C Liang and B Zhou, *Differential Geometry and General Relativity*, 2023
3. R Wald, *General Relativity*, 1984
4. V Ferrari, L Gualtieri, P Pani, *General Relativity and its Applications*, 2021
5. T Padmanabhan, *Gravitation: Foundations and Frontiers*, 2010
6. A Zee, *Einstein Gravity in a Nutshell*, 2013.
7. S Carroll, *Spacetime and Geometry: An Introduction to General Relativity*, 2003
8. Choquet-Bruhat, *Introduction to General Relativity, Black Holes, and Cosmology*, 2015
9. J Hartle, *Gravity: An Introduction to Einstein’s General Relativity*, 2003. (undergrad)
10. B Schutz, *A First Course in General Relativity 2e*, 2009. (undergrad)
11. MP Hobson, GP Efstathiou, and AN Lasenby, *General Relativity: An Introduction for Physicists*, 2006. (strong for cosmology)

For Differential Geometry

1. B O’Neill, *Semi-Riemannian Geometry*, 1983
2. S Lovett, *Differential Geometry of Manifolds 2nd Ed*, 2020

Reminders

Exam Dates

- ☐ Long Exam 1 : **October 12, 2026, 9-12 AM**
- ☐ Long Exam 2 : **December 7, 2026, 9-12 AM**
- ☐ Oral Exam : **Finals week (TBA)**

Problem Set Dates



Contents

Course Information	2
Main References	3
Reminders	4
Chapter 1: Review of Newtonian Physics	7
1.1 Introduction	7
1.1.1 Newton's Law of Gravitation	7
1.1.2 Newton's Equations of Motion	9
1.1.3 Weak Equivalence Principle	11
1.1.4 A Theory of Gravity	13
1.2 Einstein's Equivalence Principle	15
1.2.1 Inertial Frames	15
1.2.2 Universality of Free Fall	15
1.3 History of Relativity	17
1.3.1 Relativity Before Einstein	17
1.3.2 Maxwell's Results	18
1.3.3 Einstein's Contribution	19
1.3.4 From Special to General Relativity	21
Chapter 2: Special Relativity	23
Chapter 3: Aspects of Differential Geometry	24
Chapter 4: Einstein Field Equations	25
Chapter 5: Matter Models	26
Chapter 6: Weak-field Solutions	27
Chapter 7: Spherical Symmetry	28
Chapter 8: Basic Cosmology	29

List of Figures

Chapter 1

Review of Newtonian Physics

1.1 Introduction

We start with the assumption that you have no idea what General Relativity is. How do we quantify what is gravity? What even is gravity? We begin with the classical equations of motion.

To motivate ourselves, we describe three concepts introduced over the history of classical mechanics and explain how it leads to certain contradictions or problems that motivated the invention of general relativity

1.1.1 Newton's Law of Gravitation

Newton's Laws as he prescribed them is expressed as follows:

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}} \quad (1.1.1)$$

This is the familiar form of Newtonian gravity, and is perhaps one of the most recognizable equations in physics. This is what every high schooler knows and what every mentor presents, and is trivial.

We can separate the mass m experiencing the force from the mass M producing it, and write

$$\mathbf{F} = m_g \left(-\frac{GM}{r^2}\hat{\mathbf{r}} \right) \quad (1.1.2)$$

We can henceforth call m_g as the *gravitational mass* and the quantity $-\frac{GM}{r^2}\hat{\mathbf{r}}$ as the *gravitational field*

We can express it more generally by recognizing the force field is simply the negative gradient of a potential field.

$$\mathbf{F} = m_g(-\nabla\Phi) \quad (1.1.3)$$

where $\Phi = -\frac{GM}{r}$ is the gravitational potential produced by the source mass.

Thus, we can henceforth call m_g as the *gravitational charge*. It follows from the fact that

$$\text{charge} \times \text{field gradient} = \text{force} \quad (1.1.4)$$

This formulation is universal across all of physics.

Analogous to electromagnetism, where

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{\mathbf{r}} \quad (1.1.5)$$

Like the electric charge q , the gravitational charge m_g determines the strength of the interaction of the forces.

An important property that Newtonian gravitation has is that it has superposition. i.e., fields add together linearly. This is therefore also true of the gravitational potential.

For a system of point masses, the total potential field $\Phi(\mathbf{x})$ at a point $\mathbf{x}$ is given by the sum of the potentials produced by each individual mass:

$$\begin{aligned} \Phi(\mathbf{x}) &= \sum_i \Phi_i \\ &= -G \sum_i \frac{M_i}{|\mathbf{x} - \mathbf{r}_i|} \end{aligned}$$

We can extend this description from a discrete collection of point masses to a continuous mass distribution. Let the mass contained in an infinitesimal volume d^3x be

$$dm_i = \rho(t, \mathbf{x}) d^3x \quad (1.1.6)$$

Then we arrive with the continuum version of the gravitational potential field generated by a mass distribution.

$$\Phi(\mathbf{x}) = -G \int \frac{\rho(t, \mathbf{x}) d^3r}{|\mathbf{x} - \mathbf{r}|} \quad (1.1.7)$$

Here, $\mathbf{r}$ denotes the position of each source element of the mass distribution, while $\mathbf{x}$ denotes the point at which the potential is being evaluated.

We can now obtain a differential equation for the gravitational potential. Taking the Laplacian of Φ gives

$$\nabla^2 \Phi = -G \int \rho(t, \mathbf{x}) d^3r \nabla^2 \frac{1}{|\mathbf{x} - \mathbf{r}|} \quad (1.1.8)$$

Recall that this operation results to a dirac-delta function

$$\nabla^2 \frac{1}{|\mathbf{x} - \mathbf{r}|} = -4\pi\delta(\mathbf{x} - \mathbf{r}) \quad (1.1.9)$$

which relates the Laplacian of the inverse-distance potential to the Dirac delta function. Substituting this into the previous expression gives

$$\nabla^2 \Phi = 4\pi G \int \rho(t, \mathbf{x}) d^3r \delta(\mathbf{x} - \mathbf{r}) \quad (1.1.10)$$

Recall the sifting property of the Dirac delta

$$\int f(x)\delta(x-x')dx' = f(x') \quad (1.1.11)$$

From this we obtain

$$\nabla^2\Phi = 4\pi G\rho(t, \mathbf{r}) \quad (1.1.12)$$

Thus, we arrived with the fundamental field equation which governs Newtonian Gravity

Definition 1.1 : Field Equation for Newtonian Gravity

$$\nabla^2\Phi = 4\pi G\rho \quad (1.1.13)$$

The significance of a field equation is that it relates a field to its sources. In this case, the mass density ρ acts as the source of the gravitational potential field Φ .

Note the following

$$\Phi = -G \int \frac{\rho d^3r}{|\mathbf{x} - \mathbf{r}|} \iff \nabla^2\Phi = 4\pi G\rho \quad (1.1.14)$$

The two are notionally equivalent. One is a differential equation and the other is the integral form. One can switch to and from them using the Laplacian and Green's functions¹. It assigns the definition of the field, and fields assign values to every point in space.

However, this formulation presents a fundamental problem. Since the potential Φ is determined directly by the mass density ρ , any change in the mass distribution produces an immediate change in the gravitational field throughout space. There is no propagation delay or retardation. In other words, Newtonian gravity permits changes in the gravitational field to propagate instantaneously, which is incompatible with relativistic causality. This problem is one of the motivations for general relativity.

The corresponding requirement in GR is that, in the appropriate weak-field and slow-motion limit, general relativity must reduce to Newtonian gravity.

Now that we have the potential of the field that the source creates, and thus the force, how do we determine how an object moves in response to that field?

1.1.2 Newton's Equations of Motion

The quintessential equation governing the motion of classical bodies was expressed by Newton as

$$\mathbf{F} = m\mathbf{a} \quad (1.1.15)$$

Everybody and their mother knows this shit.

This states that if one sums of all the forces acting on an object, it must equal its mass multiplied by its acceleration:

$$\sum \mathbf{F} = m\mathbf{a} \quad (1.1.16)$$

¹Green's functions are essentially the inverse of a laplacian, ∇^{-2}

We can rewrite the equation as

$$\mathbf{a} = \frac{1}{m} \sum \mathbf{F} \quad (1.1.17)$$

Notice that the m term acts as a kind of resistance term², similar to the concept of inertia. This is to be expected, since Newton's first law can essentially be understood as the statement that, in the absence of a net force, an object does not change its state of motion.

Hence, we name this quantity the *inertial mass*, m_i . Inertia denotes the difficulty or resistance of an object to a change in its state of motion.

We can restate (1.1.17) with m replaced by m_i , to distinguish inertial mass from the gravitational mass introduced previously:

$$\mathbf{a} = \frac{1}{m_i} \mathbf{F} \quad (1.1.18)$$

Recall that we already have the definition of $\mathbf{F}$ from (1.1.3). Thus, to calculate the motion of an object under the influence of gravity, we simply calculate its acceleration:

$$\mathbf{a} = \frac{m_g}{m_i} (-\nabla\Phi) \quad (1.1.19)$$

Notice that this is similar in form to the equation of motion for a charged particle in an electric field.

$$\mathbf{a} = \frac{q}{m_i} \mathbf{E} \quad (1.1.20)$$

where m_i is the inertial mass, the measure of an object's resistance to changes in its motion. In both cases, the coupling to the field determines the force, while the inertial mass determines the resulting acceleration.

However, for calculations involving Newtonian gravity, the ratio of the gravitational charge and inertial mass $\frac{m_g}{m_i}$ is assumed to be 1. Thus, for gravity, the acceleration of a body is expressed as

$$\mathbf{a} = -\nabla\Phi. \quad (1.1.21)$$

This assumes that

$$m_g = m_i, \quad (1.1.22)$$

This feels unnatural and unexpected. Why should it be the case that one force acts differently from other forces in terms of its charge carriers? Mass is a fundamental property that affects other forces, there's no immediate evidence why mass is gravity's charge carrier.

There is no *a priori* reason for the gravitational charge and inertial mass to be equivalent. They play fundamentally different roles in the theory. It is analogous to assuming that an object's coupling to some other field must be equal to its inertial mass, like an electron's electromagnetic properties being dependent on mass.

Nevertheless, for gravity this equality is accepted experimental fact without explanation.

²Recall Ohm's Law $V = IR$.

1.1.3 Weak Equivalence Principle

The assumption that

$$m_i = m_g \quad (1.1.23)$$

is called the *weak equivalence principle*.

It follows that, for objects only affected by gravity, the equation of motion is simply

$$\mathbf{a} = -\nabla\Phi. \quad (1.1.24)$$

That is, any test mass placed in the same gravitational field experiences the same acceleration, regardless of its mass or composition. The nature of the object itself does not appear in the equation of motion. The field prescribes the acceleration, the test body merely responds to it.

One could therefore remove the test object m and nothing about the gravitational field would change. In Newtonian mechanics, this is treated as a universal fact: the gravitational field exists independently of the particular body used to probe it.

However, this raises a deeper question.

One could argue that the stage for the motion of an object is the field itself. If we remove the object from some local region, the field remains.

By the same logic, if we remove the field itself, what remains? In Newtonian mechanics, space $\mathbb{R}^3$ remains as the stage on which the field and matter exist.

The influence of gravity has nothing to do with the nature of the test object itself. The gravitational field does not change when the test object is removed, and its motion depends only on the field at its location.

One might ask: if we remove all mass, would there not be no gravity? The answer is that we must distinguish the sources of the field from the test body. We are considering local interactions. Mass may exist somewhere outside the local region of interest and still generate a gravitational field within it. Given the local potential Φ , we can determine information about its sources, but the test body itself is not one of those sources.

Thus,

$$\mathbf{a} = -\nabla\Phi \quad (1.1.25)$$

is treated as inertial motion in a gravitational field.

In the stage analogy, one could think of $-\nabla\Phi$ as living on the stage of space $\mathbb{R}^3$. The field exists on space and determines how objects move through it. In some fields such as particle physics, that is the case, they treat gravity as a special field which acts on every particle and every particle responds to in the same way

However, in GR, $-\nabla\Phi$ is no longer treated as a field sitting on top of the stage. The gravitational field *is* the stage. The space time is what the rest of physics takes place in.

The strong equivalence principle extends this idea: locally, the laws of physics in a freely falling frame are the same as those of an inertial frame in the absence of gravity. In other words, locally, gravity can be transformed away by choosing an appropriate freely falling frame.

Inertial motion is therefore determined by the structure of spacetime.

Physics takes place on the stage called spacetime, and GR governs that space time. The spacetime structure remains even after the particular body used to probe it is removed.

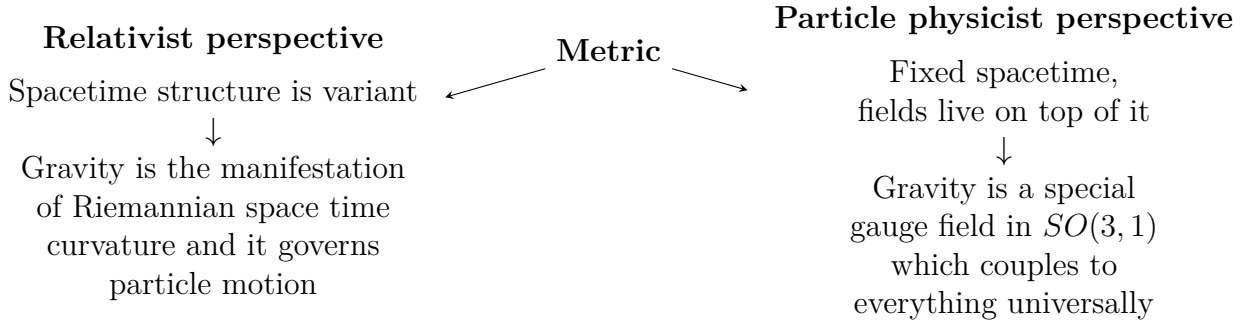
In GR, instead of the gravitational potential field Φ , we have the metric:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \quad (1.1.26)$$

The metric determines the geometry of spacetime and, consequently, the inertial trajectories of objects.

In particle physics, one generally begins with a fixed flat spacetime $\mathbb{R}^4$ and defines fields upon that background. Gravity can then be treated as a field on this background, rather than as the geometry of spacetime itself. This approach is sound as a field-theoretic formulation and provides a natural route toward quantizing gravity and describing gravitons, but it is ugly and inelegant. Regardless, you can bootstrap yourself from there and arrive on the EFE

The distinction of perspectives can be summarized as follows:



Regardless of the formalism, the different approaches must reproduce the same classical predictions. The geometric formulation of GR ultimately leads to the Einstein field equations.

1.1.4 A Theory of Gravity

General Relativity, to be a theory of gravity, must explain gravitational phenomena.

For example, Newtonian gravity became a theory of gravity because it explained the results Kepler had found. Kepler described the motion of the planets and discovered that their orbits were elliptical using data analysis, but he did not have an underlying physical theory explaining *why* they followed these trajectories.

Newton instead derived the observed behavior from first principles. His laws of motion and universal gravitation explained Kepler's laws and provided a general framework for celestial mechanics, and ended up predicting the whole of kinematics and dynamics.

As such, GR as a theory of gravity must be able to explain observed gravitational phenomena: relativistic corrections to celestial mechanics, gravitational waves, black holes, gravitational collapse, and, on the largest scales, cosmology.

Perhaps the grandest application of GR is cosmology. Newtonian gravity is fundamentally a local theory applied within a fixed space and time, while GR allows the spacetime itself to be dynamical. This makes it possible to formulate a theory describing the universe as a whole.

DISTINCTION: For something to be a theory of gravity, what must it possess?

1. Dynamical variables

The theory must have variables describing the gravitational field in which objects move.

$$\begin{array}{ll} \text{Newton:} & \Phi \\ \text{GR:} & g_{\mu\nu} \end{array}$$

2. A law describing how the field affects objects

The theory must describe how its dynamical variables determine the motion of objects.

For Newtonian gravity:

$$\mathbf{a} = -\nabla\Phi. \quad (1.1.27)$$

For GR, freely falling objects follow geodesics:

$$\frac{d^2x^\sigma}{d\lambda^2} + \Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0, \quad (1.1.28)$$

or, using the four-velocity u^μ ,

$$u^\mu \nabla_\mu u^\nu = 0. \quad (1.1.29)$$

3. A law describing how objects produce the dynamical variable

Finally, the theory must tell us how matter produces the gravitational field.

For Newtonian gravity:

$$\nabla^2\Phi = 4\pi G\rho. \quad (1.1.30)$$

For GR:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi GT_{\mu\nu}. \quad (1.1.31)$$

Since the two formalisms attempt to explain the same phenomenon, we demand that the GR formulation reduces to the Newtonian formulation in weak fields.

These three ingredients are the basic structure of a theory of gravity: a dynamical variable, a law governing how it affects matter, and a field equation governing how matter produces it.

GR is inherently nonlinear. Going from the field to the motion of objects is already more complicated than in Newtonian gravity, but the field equation itself is also nonlinear: the metric determines the curvature, while the curvature determines the metric.

Solving the Einstein field equations for realistic systems is therefore often extremely difficult. In many cases, exact solutions cannot be obtained, and the equations must be solved numerically. Careers have been made doing this. This is the domain of *numerical relativity*.

In the next chapters, we will explore how gravity can be explained by general relativity.

1.2 Einstein's Equivalence Principle

1.2.1 Inertial Frames

The Weinberg definition is that, at every point in spacetime, i.e. at every (t, x, y, z) in a gravitational field, it is possible to choose a locally inertial frame, i.e. a freely falling coordinate system, such that the “laws of nature” take the same form as they do in an unaccelerated Cartesian coordinate system in the absence of gravity.

In other words, fields that are accelerating are locally indistinguishable from gravity. This is the basic idea behind the equivalence principle.

Gravity can therefore be regarded as a fictitious force, because it can be removed by a coordinate transformation. It is there only because we are using the wrong coordinate system.

The equivalence principle states that it is always possible to remove gravity locally. Therefore, gravity is a fictitious force” that can be “removed” when describing the local field in an experiment.

The important difference between gravity and other fictitious forces is that gravity can only be removed locally, while other fictitious forces can be removed globally. The reason is that a gravitational field has spatial variation. Even after transforming away the gravitational acceleration at one point, neighboring points may experience slightly different accelerations.

We say that, regardless of the mass of the object, it does not take part in determining its own gravitational acceleration. The gravitational acceleration is universal.

The measurement of differences between these gravitational fields is called *tidal* measurement, or simply the measurement of tides. These tidal effects cannot be transformed away over an extended region.

1.2.2 Universality of Free Fall

The usual way this is expressed is to imagine a person in a box conducting a free-fall experiment. Suppose they perform some experiment in which they can measure the acceleration of objects.

For an object o_1 , they measure

$$\mathbf{a}_1 = \frac{m_{1g}}{m_{1i}} \mathbf{g} \quad (1.2.1)$$

and for a second object o_2 , they can measure

$$\mathbf{a}_2 = \frac{m_{2g}}{m_{2i}} \mathbf{g}. \quad (1.2.2)$$

The universality of free fall is the statement that

$$\mathbf{a}_1 = \mathbf{a}_2 \quad (1.2.3)$$

under the same circumstances, regardless of the nature or mass of the objects. This is called the *universality of free fall*.

In Newtonian language, this corresponds to the equality of gravitational and inertial mass. The important point for the equivalence principle is that all sufficiently small freely falling objects respond to the gravitational field in the same way.

In a simple example, we can write the gravitational field as a non-inertial frame:

$$\ddot{y} = -g \quad (1.2.4)$$

We can remove gravity, however, via certain transformations. Let

$$z = y - \frac{1}{2}gt^2 \quad (1.2.5)$$

This results in

$$\ddot{z} = 0. \quad (1.2.6)$$

This is an inertial coordinate system, since there is no perceived acceleration. The point is that gravity can be removed locally by choosing an appropriate freely falling coordinate system.

The person in the box cannot tell whether they are accelerating upward with constant acceleration g , or whether they are freely falling under the influence of gravity. In a sufficiently small region, the two situations are experimentally indistinguishable.

Gravity is therefore, locally, simply the manifestation of being in the wrong frame.

Inertial means the absence of gravitational acceleration in the local frame. More precisely, an inertial frame is one in which freely falling objects move without coordinate acceleration.

Weak, Strong, and Einstein Equivalence Principles We have the weak equivalence principle, where the “laws of nature” are understood primarily as the laws of mechanics and the universality of free fall.

For the strong equivalence principle, one extends the statement to gravitationally self-interacting or strongly gravitating systems, so that the equivalence applies more broadly than to test particles alone.

The Einstein equivalence principle goes further: “laws of nature” means all nongravitational laws of nature. It is one of the conceptual foundations of general relativity.

In any theory of gravity, one can attempt to remove g locally by an appropriate choice of coordinates. What makes general relativity special is the geometric interpretation of what remains after the local gravitational acceleration has been removed.

It means that gravity is genuinely fictitious locally if no experiment confined to that sufficiently small region can tell that we are in a gravitational field. If one can remove gravity, then what remains should be the stage itself: the underlying spacetime structure and its geometry.

1.3 History of Relativity

The notion of inertiality plays a major role in relativity. Let us first explain relativity in its historical context.

We begin with special relativity and Newtonian gravity. The combination of these ideas eventually leads us toward general relativity, where gravity is incorporated into the structure of spacetime itself.

1.3.1 Relativity Before Einstein

What is relativity? It is an old notion that existed even by the time of Newton. The essential idea is that there should exist observers for which the laws of nature take the same form. Regardless of where the observer is or how they move within the appropriate class of frames, the laws of nature should be invariant.

This is the equivalence of *inertial observers*: the laws of nature take the same form for all inertial observers.

Aristotle did not have this notion. He believed that there was a special place toward which objects naturally moved: the center of the universe, identified in practice with the center of the Earth. One could therefore imagine that this special region defined the natural frame in which the laws of nature took their proper form.

It made sense within that picture to express physics relative to the center of the Earth, since all other frames could be considered as moving relative to this preferred location. For example, one could imagine waves in a pond as being fixed relative to the pond itself, with the pond providing a preferred rest frame.

Galileo rejected this notion by positing that there is not a single preferred inertial state. Instead, there is a class of observers for whom the laws of mechanics are equally valid. These observers move uniformly relative to one another.

This is called *Galilean relativity*. There is a class of observers which all move relative to each other with uniform velocities.

The transformation between two such frames can be written as

$$t' = t \tag{1.3.1}$$

$$\mathbf{x}' = \mathbf{x} - \mathbf{v}t \tag{1.3.2}$$

This relation explains how one can switch to the frame of reference of a second observer moving at velocity $\mathbf{v}$ relative to the first.

In that case, one has the following invariance of Newtonian mechanics:

$$\mathbf{F} = m\mathbf{a} \implies \mathbf{F}' = m\mathbf{a}'. \tag{1.3.3}$$

Newtonian theory therefore satisfies Galilean relativity: it is invariant under Galilean transformations.

From the development of Newtonian mechanics onward, classical mechanics became the fundamental framework of physics. For several centuries, this was the dominant picture of physical law.

1.3.2 Maxwell's Results

For roughly three centuries, Newtonian mechanics was the jewel of physics. Then electromagnetism broke this picture.

In the nineteenth century, Maxwell showed that Maxwell's equations are not Galilean invariant. They therefore do not satisfy Galilean relativity in the same way that Newton's equations do. This challenged centuries of orthodoxy.

The response of academia was initially to deny that Maxwell's equations were fundamental in the same sense as Newtonian mechanics. One possibility was that Maxwell's equations were only correct in the rest frame of some physical medium, and that medium was called the *aether*.

The aether was imagined as an invented medium through which electromagnetic waves propagated. In this picture, Maxwell's equations described electromagnetic waves in the same broad conceptual sense that waves in a pond are described relative to the pond.

Light would then move at speed c only in the rest frame of the aether.

This led naturally to attempts to detect the Earth's motion through the aether. The Michelson–Morley experiment was a null experiment: the expected difference in the measured speed of light was not observed, and the result indicated that c was the same in the different experimental directions.

People fought against this result, as they should have. The natural response to an experimental discrepancy is to try to preserve a successful theory if possible and search for a physical explanation for the new result.

Ad Hoc Solutions to the Aether Problem. One proposed solution was Lorentz contraction. Objects would contract in the direction of motion through the aether, although there was initially no satisfactory physical explanation for why they should do so. It was essentially an ad hoc solution introduced to maintain consistency with the null result.

Another proposed idea was *ether dragging*. Imagine that the ether is not a fixed frame, but behaves something like molasses or syrup, so that the Earth in motion drags the ether around it. The Earth would then remain locally at rest with respect to the ether, and there would consequently be no observed change in c .

The debates went on for decades. People had built careers around various versions of these theories, and many ad hoc solutions were introduced to preserve the compatibility of Maxwell's theory with Galilean relativity.

There were dozens of theories introduced in the hope of making Maxwell's equations consistent with the Galilean framework.

1.3.3 Einstein's Contribution

Einstein came in 1905 with his *annus mirabilis* papers.

The three major papers associated with this period gave rise to major developments in quantum theory, statistical mechanics, and special relativity. Einstein developed these ideas all before turning 24.

Everyone had been focused on finding a way to mesh Galilean relativity with Maxwell's equations. The common strategy was to deny the fundamentality of Maxwell's equations by claiming that they only worked in the preferred frame of the aether.

Einstein instead reversed the argument.

Einstein posited that Maxwell's equations are fundamental, while Newtonian mechanics is not fundamental. Newtonian mechanics was a theory that had been successful for three centuries, but its enormous historical success did not make it fundamentally immune to revision.

His solution is:

- adopt c as universal;
- accept Maxwell's equations as fundamental.

As such, the equations governing Galilean relativity must be dropped so that c remains constant. By consequence, one also has to give up or modify Newtonian mechanics and stop treating it as fundamentally exact.

One has to divorce oneself from the notion that Newtonian mechanics is fundamental.

The notion of universal observers is still true: there is a class of observers for whom the laws of motion retain the same form. However, to reconcile this with the invariance of c , one must change the transformation law between observers.

This leads to the unification of space and time and to the Lorentz transformations.

Lorentz Transformations and Covariance Maxwell's theory is relativistic. It is relativistically covariant: under the appropriate relativistic transformations, its mathematical form does not change.

The task is therefore to find the correct transformation laws that keep Maxwell's equations invariant.

The Poincaré-Lorentz transformations provide the appropriate transformations:

$$ct' = \gamma(ct - \beta x) \tag{1.3.4}$$

$$x' = \gamma(x - \beta ct) \tag{1.3.5}$$

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} \tag{1.3.6}$$

$$\beta = \frac{v}{c}. \tag{1.3.7}$$

The project, according to Einstein, is to rewrite all of physics so that it is in “relativistically covariant” form: the equations should be invariant in form from one inertial observer to another.

The form of these transformations is

$$x^{\mu'} = \Lambda^\mu_{\nu} x^\nu. \quad (1.3.8)$$

Where

$$x^\mu = (ct, x^1, x^2, x^3) \quad (1.3.9)$$

and

$$\Lambda^\mu_{\nu} = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1.3.10)$$

Now, the force equation becomes

$$\frac{d\mathbf{p}}{dt} = \mathbf{F} \implies \frac{dp^\mu}{d\tau} = F^\mu. \quad (1.3.11)$$

The second equation reduces to the first in the appropriate nonrelativistic, low-velocity limit.

How About Gravity? Gravity is simply

$$\nabla^2 \Phi = 4\pi G \rho. \quad (1.3.12)$$

This is not relativistically covariant, since it is written as a theory involving only space and does not incorporate space and time into a unified relativistic structure.

One can try to modify it so that

$$\square \Phi = 4\pi G T^\mu_{\mu}. \quad (1.3.13)$$

This is known as Nordström gravity. But this theory is incorrect as a theory of gravity because it does not predict the bending of light.

However, Einstein was bothered by the equivalence principle. Why is there an equivalence of falling? Why does gravity have this peculiar property that all objects fall in the same way?

This question leads to the deeper geometric interpretation of gravity.

Minkowski Spacetime It was Minkowski in 1907 who placed these transformations in their modern geometric form.

One can make sense of special relativity by imagining space and time as a single structure: *spacetime*. This spacetime possesses a metric, now known as the Minkowski metric,

$$\eta_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1.3.14)$$

and the transformations that preserve the Minkowski metric are the correct relativistic transformations.

The key conceptual step is that space and time are no longer treated as two completely separate structures. They form a single spacetime geometry, and inertial observers correspond to the natural geometric structure of this spacetime.

1.3.4 From Special to General Relativity

In 1915, Einstein formalized his theory of General Relativity, where he formalized the notion of spacetime and recognized that spacetime need not be described only by the Minkowski metric.

The replacement is

$$\eta_{\mu\nu} \rightarrow g_{\mu\nu}. \quad (1.3.15)$$

The Minkowski metric is therefore promoted from the fixed metric of special relativity to the more general spacetime metric $g_{\mu\nu}$, which may vary from place to place.

Einstein expanded the notion of inertial observers to all observers in free fall within spacetime. Inertial observers are therefore no longer merely observers moving with uniform velocity relative to one another. They are also freely falling observers, following geodesics of the spacetime geometry.

He got the basic idea relatively early, but obtaining the field equations took much longer.

Relativity is very, very old, and the important thing Einstein did was expand it into a general theory in which the geometric structure itself becomes dynamical.

Minkowski's relativity is special relativity, represented by η , called special because it is restricted to a class of observers moving uniformly relative to one another.

The general form, represented by g , becomes general relativity, with gravity incorporated into the geometry of spacetime.

Machian and Relational Ideas There is a distinction here involving the idea of being *Machian*. A Machian picture suggests that everything should be relational and that nothing should be absolute. In essence, Einstein was a Machian, but the theory that he founded is fundamentally not Machian.

General relativity has a deeply relational character, although it is not simply Machian in every possible sense. The geometry of spacetime itself becomes part of the physical description rather than serving merely as an immutable background stage.

Relativity Is More General Than Light There is a notion that light is inherently tied to relativity. However, even if Maxwell's theory were somehow removed or replaced, the principle of relativity itself would remain.

Light has nothing fundamentally to do with the concept of relativity. Its role is historical: Maxwell's equations were what forced physicists to reconsider Galilean relativity and search for a new transformation law, but the principle of relativity is conceptually distinct from electromagnetism.

Relativity is a metaprinciple. Maxwell inspired the modern development of relativity, but Maxwell's theory is not the conceptual foundation of the principle itself.

Light is therefore not intrinsically tied to relativity. One can derive general relativity without making light the starting point, and obtain the same gravitational equations.

The next lecture goes on to deriving general relativity while making no fundamental use of light, yet arrives at the same equations.

The Role of c The only essential role of Maxwell's theory in this historical development is to establish c as a universal constant that characterizes the structure of spacetime.

We can approach relativity with different values of the invariant speed. If we assign

$$c = \infty, \tag{1.3.16}$$

we recover Galilean relativity.

If we assign

$$c = 0, \tag{1.3.17}$$

we obtain Carrollian relativity.

And if we assign c to be an arbitrary finite number between 0 and ∞ ,

$$c \in (0, \infty), \tag{1.3.18}$$

we obtain the relativistic structure associated with special relativity, and ultimately the framework in which general relativity is formulated. We will tackle this in the next chapter.

Chapter 2

Special Relativity

Chapter 3

Aspects of Differential Geometry

Chapter 4

Einstein Field Equations

Chapter 5

Matter Models

Chapter 6

Weak-field Solutions

Chapter 7

Spherical Symmetry

Chapter 8

Basic Cosmology